

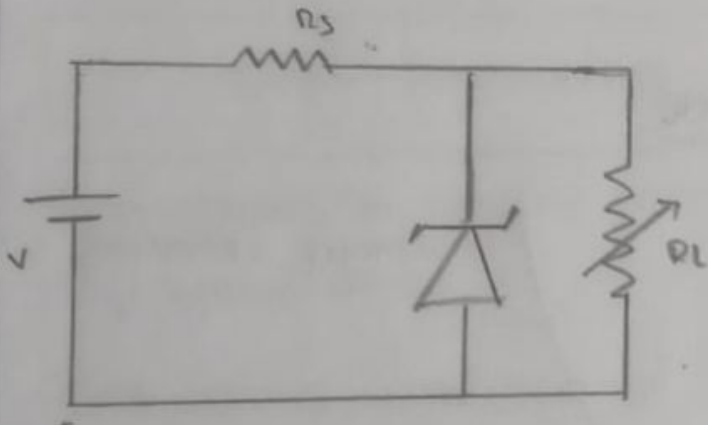
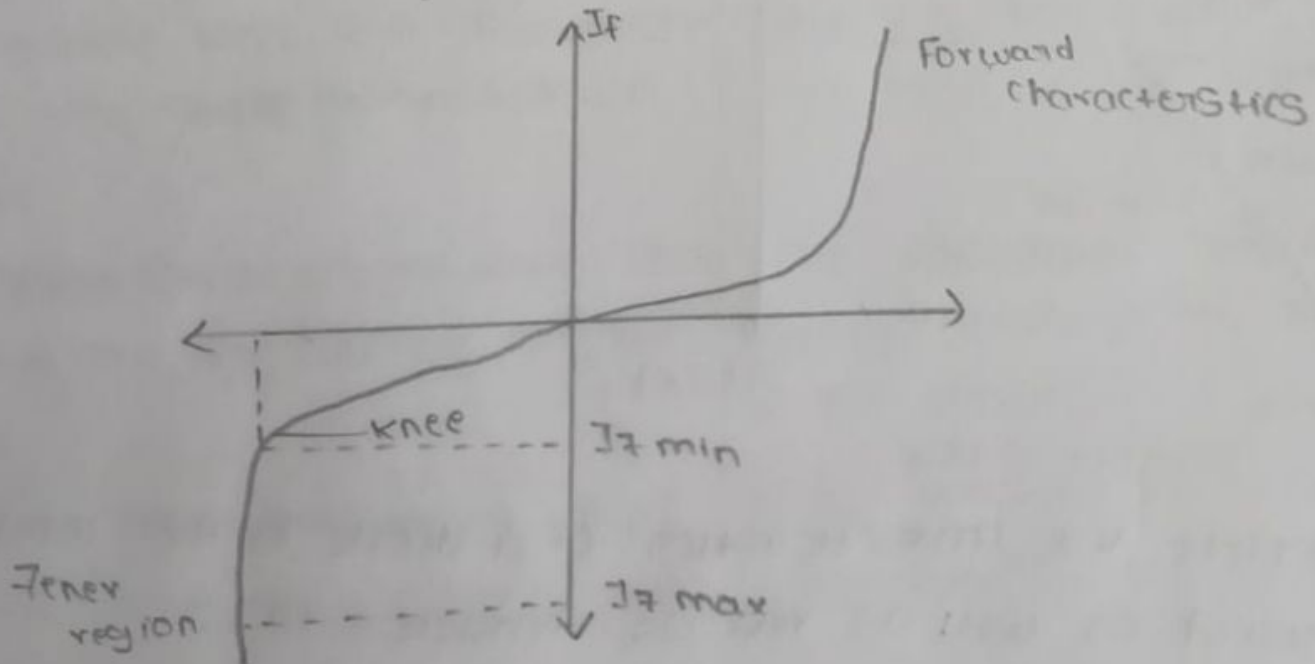
2 marks① Applications of PN Junction diode:-

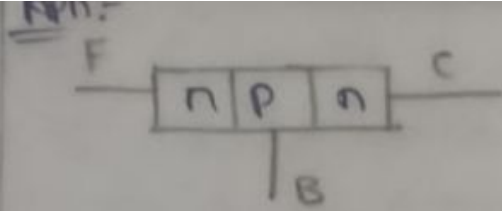
- 1) Rectification - used in rectifiers to convert AC into DC.
- 2) Switching - Acts as an electronic switches in circuits
- 3) clipping & clamping circuits - used to modify wave form shapes.
- 4) Detection / demodulation - used in radio receivers to extract signals

② Breakdowns in PN Junction diode:-

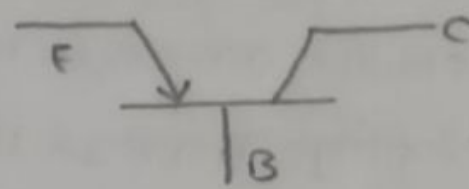
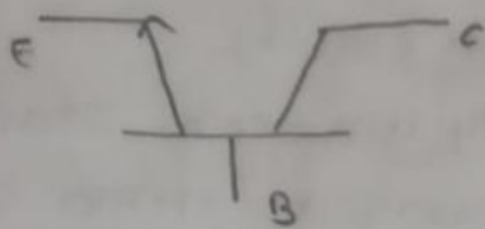
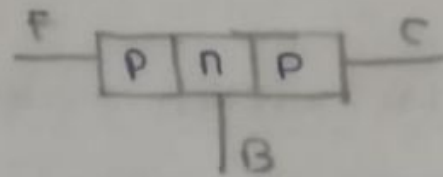
- 1) Zener breakdown - occurs at low reverse voltage due to strong electric field.
- 2) Avalanche Breakdown - occurs at high reverse voltage due to collision of charge carriers.

③

 R_L - load resistor R_s - series resistor.④ characteristic of Zener diode:-⑤ transistor symbol:- $\Rightarrow$



PNP:

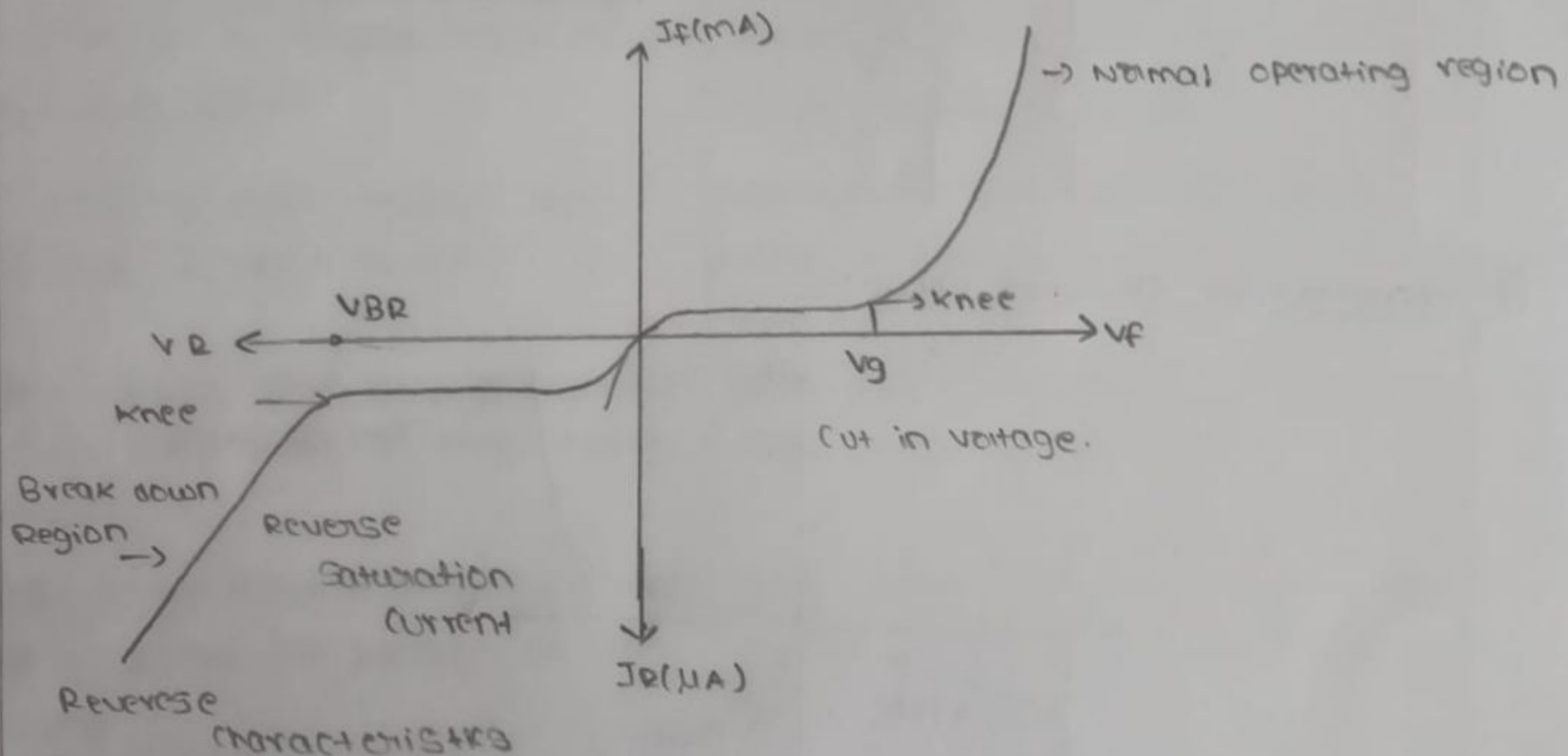


Various configuration of transistor:-

- 1) common Base (CB) :: Base is common to input and output.
- 2) common Emitter (CE) :: Emitter is common
- 3) common collector (CC) :: collector is common, also called emitter follower

10 marks

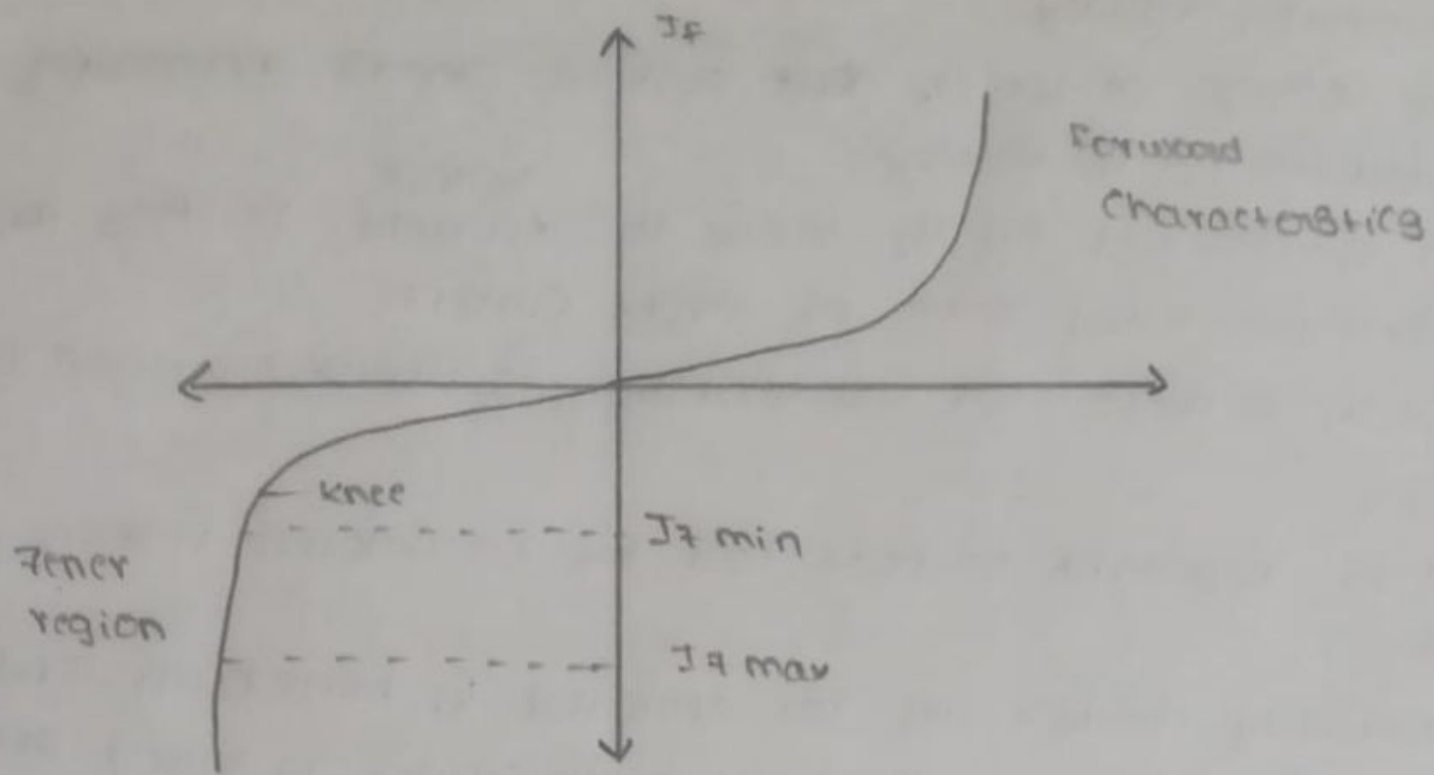
③ V-I characteristics of PN Junction diode:-



The complete V-I characteristics of a diode is the combinations of its forward as well as reverse characteristics.

In forward characteristics, it is seen that initially forward current is small along as the bias voltage is less than that the barrier potential.

At a certain voltage close to the barrier potential, current



1) Forward Bias region:

- When the diode is forward biased (+ve voltage applied to p-side) current starts flowing.
- After the threshold voltage ($\sim 0.7V$ for Silicon), current increases rapidly.
- The curve rises steeply in the +ve voltage region.

2) Reverse Bias region:-

- When reverse voltage is applied, only a small leakage current flows.
- Current remains almost constant and very low.
- This region is before breakdown (safe operating region).

3) Zener region (Breakdown):-

- When reverse voltage reaches Zener voltage, breakdown occurs.
 - Current increases sharply.
 - Voltage across the diode remains almost constant.
- $\therefore$ This is the working region of Zener diode for voltage regulation.

Key points:-

- * Zener diode is designed specially to work in reverse breakdown region.
- * Maintains constant voltage.
- * Widely used in voltage regulation.

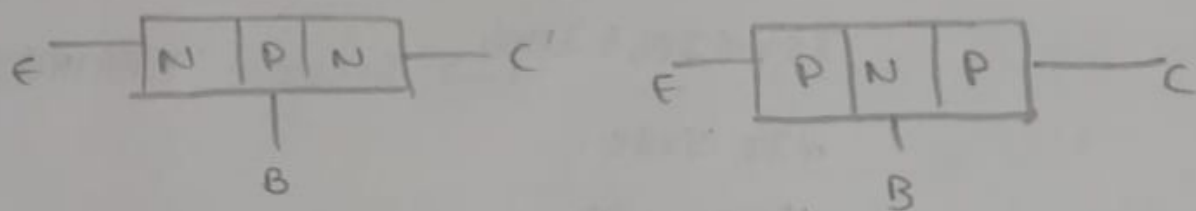
device formed by the joining two PN Junction. It can be

- NPN transistor
- PNP transistor.

Construction:-

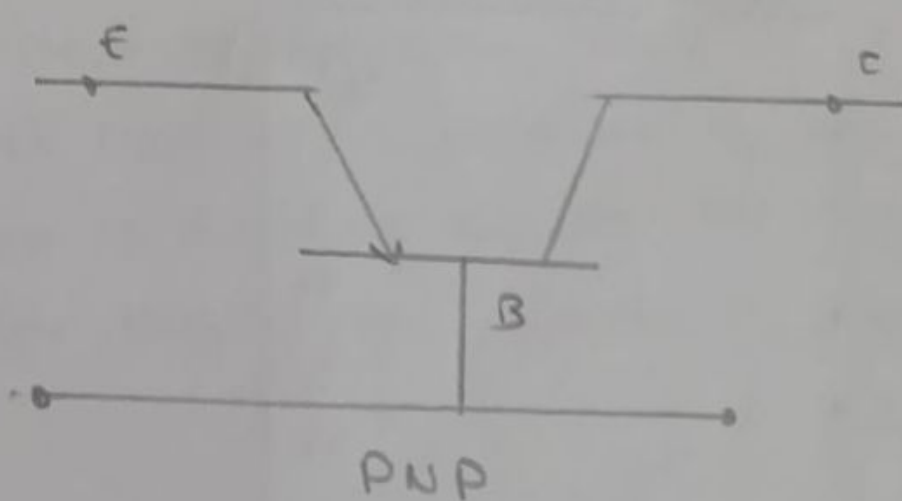
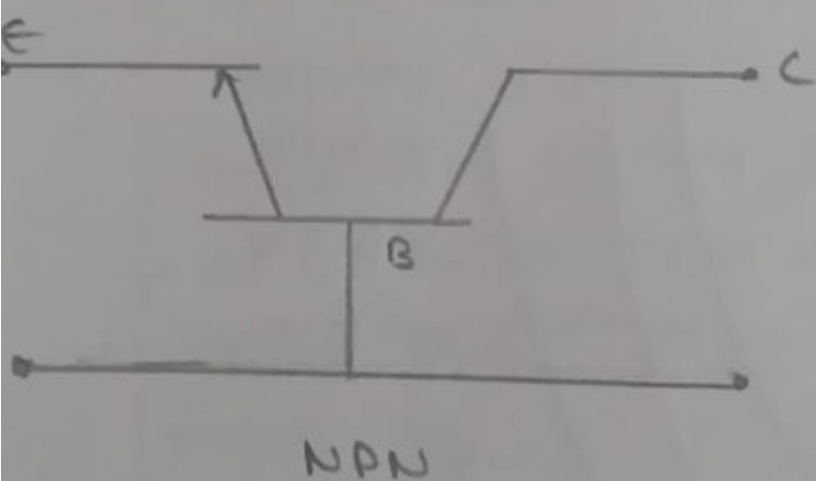
The BJT consists of a Silicon (or germanium) crystal in which a thin layer of N-type silicon is sandwiched b/w two layers of P-type silicon. This is referred as PNP.

Alternately in a NPN transistor, a layer of P-type material is sandwiched b/w two layers of N-type material. The two types of the BJT



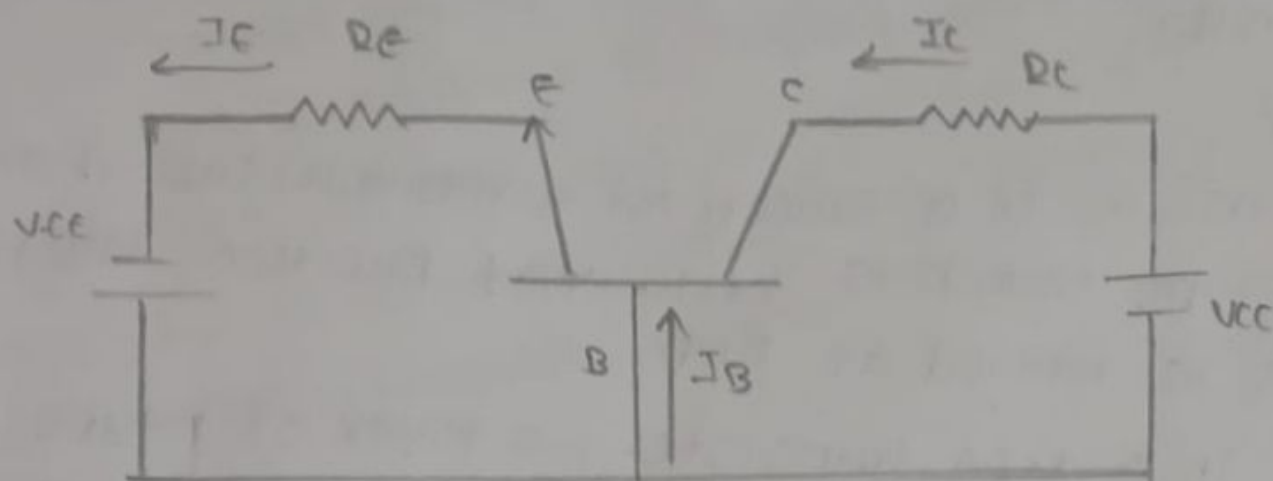
The symbolic representation of the two types of BJT is the three portions of the transistor are emitter, Base, collector represents as E, B, C respectively.

The arrow on the emitter represents the direction of current flow when the EB is forward biased.



CB configuration:-

In common Base configuration, the base is grounded and it is used as the common terminal for both input and output.



$$I_E = I_B + I_C$$

$$I_C = \alpha I_E + I_{CO}$$

$$I_C = \alpha I_E + I_{BCO} \quad \rightarrow \text{Reverse breakage current}$$

$$\alpha I_E \gg I_C$$

$$I_C \approx \alpha I_E$$

$$\alpha = \frac{I_C}{I_E} = \frac{\text{Output current}}{\text{Input current}}$$

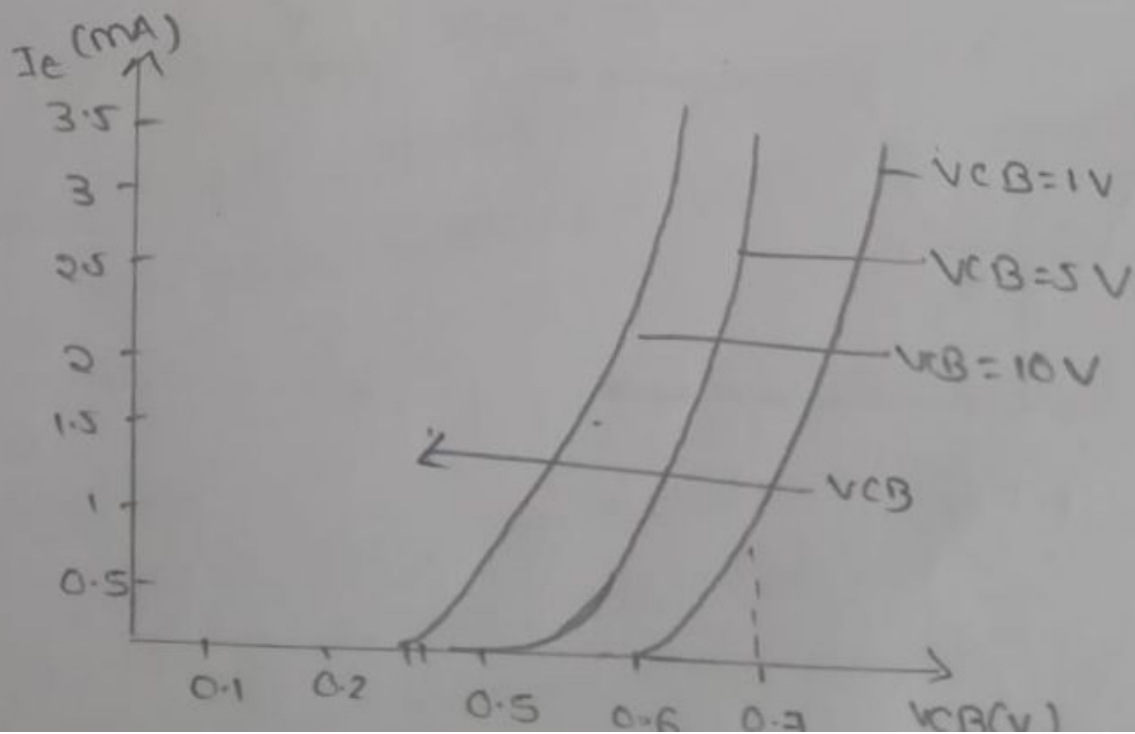
α = amplification factor (or) gain factor.

$$I_B = I_E - I_C$$

$$I_B = I_E - \alpha I_E$$

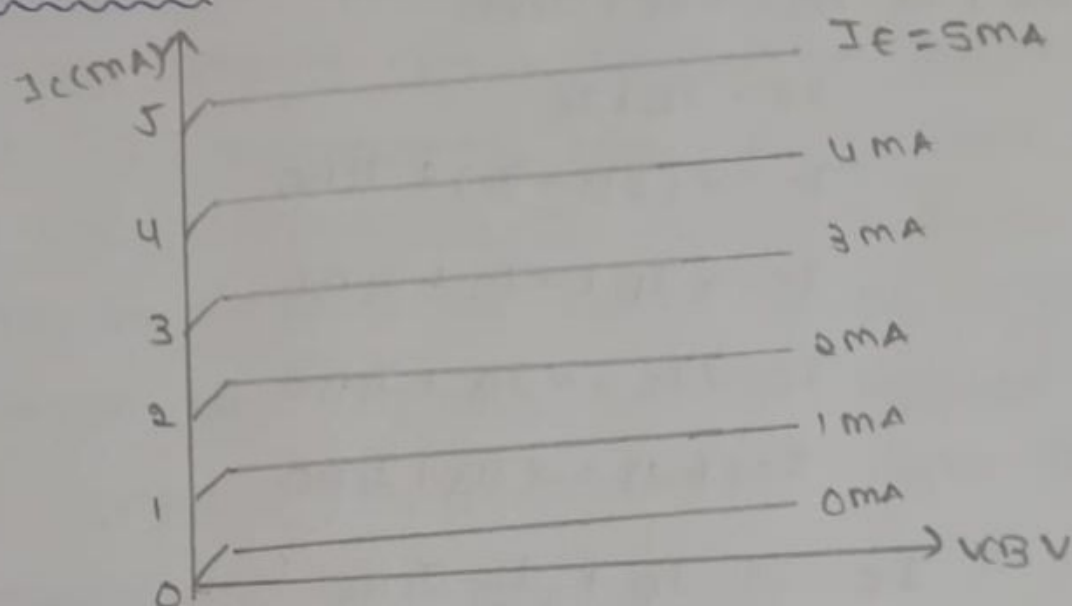
$$I_B = I_E (1 - \alpha)$$

Input characteristics:-



- It is defined as the characteristic curve drawn between input voltage to input current where as output voltage is constant.
- To determine the I_c , the collector base voltage V_{CB} is kept constant at zero and emitter current I_E is increased from zero by increasing the V_{EB} . This is repeated for higher values of V_{CB} .
- A curve is drawn b/w I_E and V_{EB} at constant V_{CB} .
- When V_{CE} is zero EB is forward bias, so it behaves as a diode so that emitter current increases rapidly.

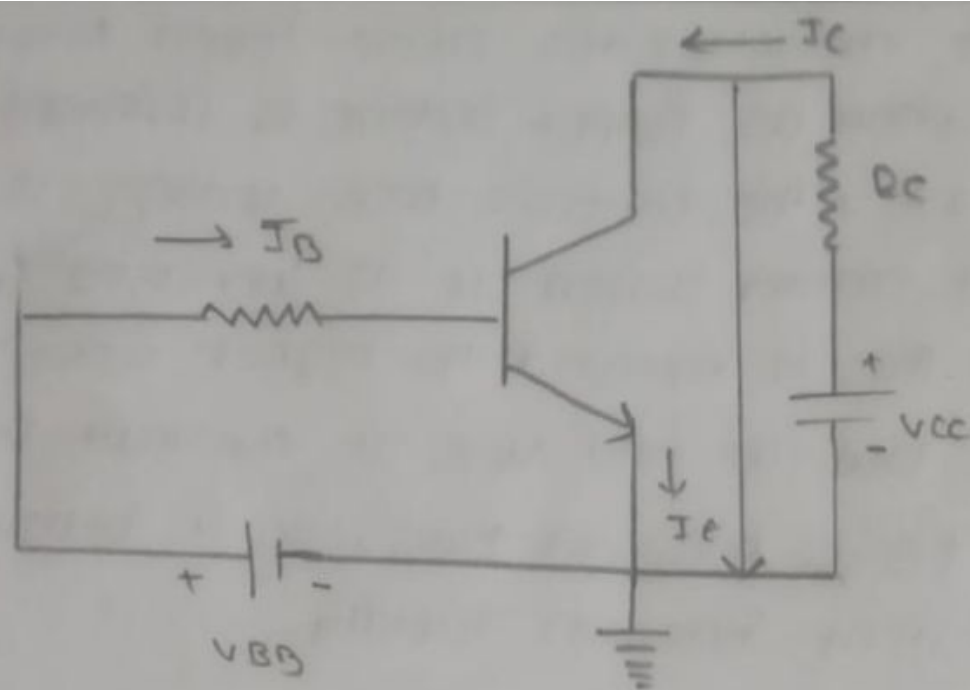
Output characteristics:-



- It is defined as the characteristic curve drawn b/w output voltage to output current where as input current is constant.
- To determine O_c , the emitter current I_E is kept constant at zero and collector current I_c is increased from zero by increasing V_{CB} . This is repeated for higher values of I_E .
- From the graph it is seen that for constant value of I_E , I_c is independent on V_{CB} and the curves is parallel to V_{CB} axis.
- As the emitter base junction is forward biased the majority charge like e^- from emitter region are injected to the base region.

8) CE Configuration:-

In a CE configuration, the emitter terminal of transistor is common to the both the input and output circuit. it is also known as grounded emitter configuration.



$$\text{W.K.T } \Rightarrow I_C = \alpha I_E + I_{CBO}$$

$$I_E = I_B + I_C$$

$$I_C = \alpha (I_B + I_C) + I_{CBO}$$

$$I_C = \alpha I_B + \alpha I_C + I_{CBO}$$

$$I_C - \alpha I_C = \alpha I_B + I_{CBO}$$

$$I_C (1 - \alpha) = \alpha I_B + I_{CBO}$$

$$I_C = \frac{\alpha}{1 - \alpha} I_B + \frac{1}{1 - \alpha} I_{CBO}$$

$$\beta = \frac{\alpha}{1 - \alpha}$$

$$I_C = \beta I_B + (\beta + 1) I_{CBO} \quad \xrightarrow{\quad} I_{CEO}$$

Case : 1

$$\alpha = 0.98$$

~~$$\alpha = 0.98$$~~

$$\beta = 1 - 0.98$$

$$\beta = 49$$

Case : 2

$$\alpha = 0.95$$

$$\beta = 1 - 0.95$$

$$\beta =$$

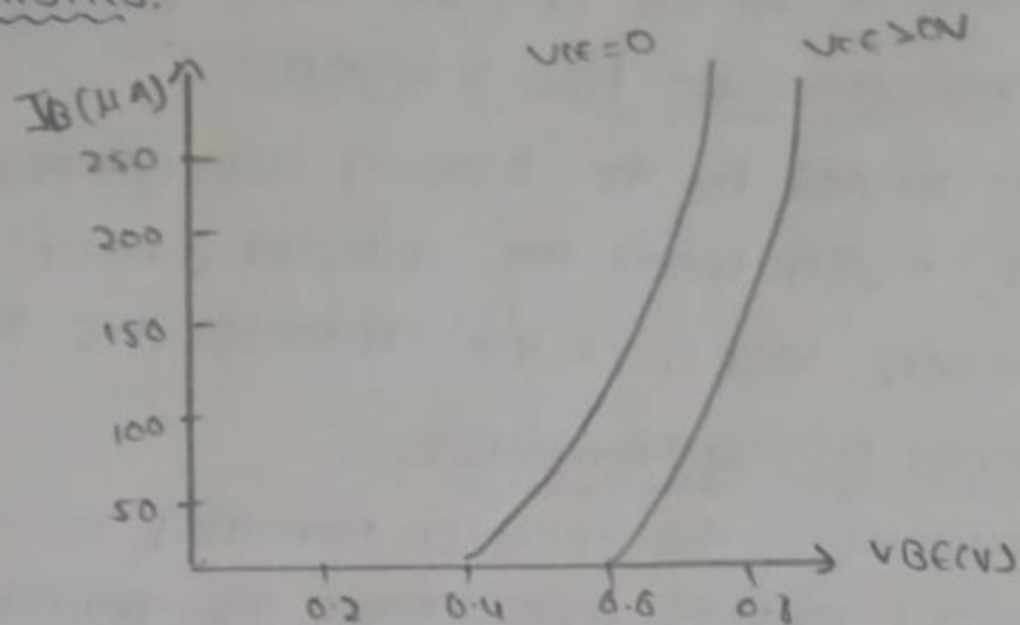
$$\alpha < \beta$$

$$I_C = \beta I_B + I_{CEO}$$

$$I_C = \beta I_B$$

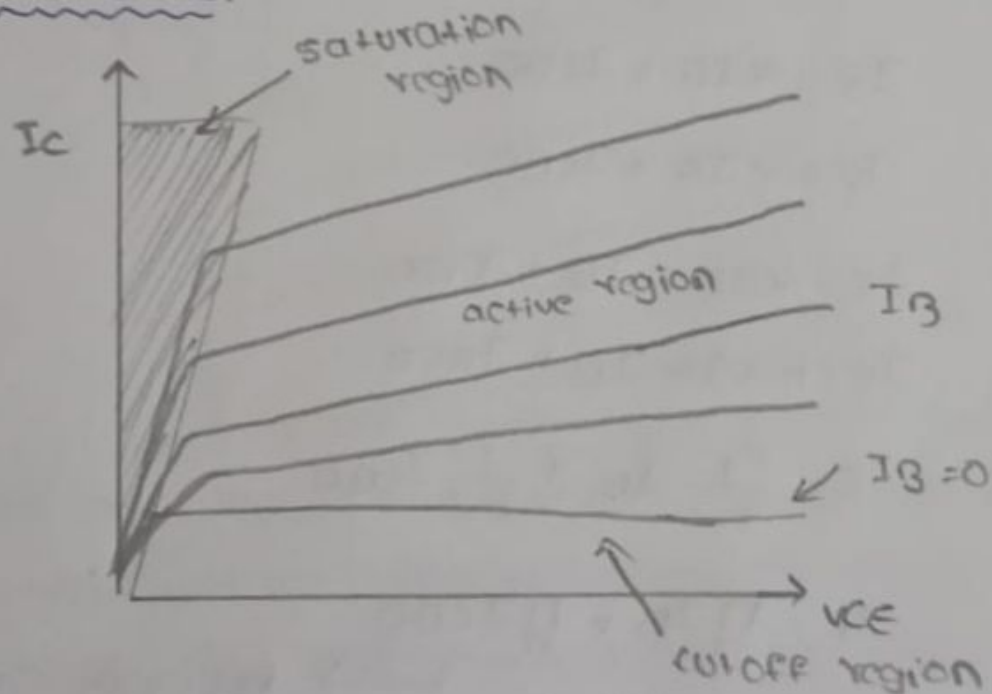
$$\beta = I_C / I_B$$

Input characteristics:-



- It is defined as the characteristic curve drawn b/w input current to input voltage whereas output voltage is constant.
- To determine i/c, the collector base voltage V_{CB} is kept constant at zero and base current I_B is increased from zero by increasing V_{BE} . This is repeated for higher values of V_{CE} .
- A curve is drawn b/w I_B and V_{BE} at constant V_{CB} .
- Base width increases, so curve moves right as V_{CE} increases.

Output characteristics:-



- It is defined as the characteristic curve drawn b/w output voltage to output current whereas input current is constant.
- To determine o/c, the base current I_B is kept at constant at zero and collector current I_C is increased by increasing V_{CE} . This is repeated for higher values of I_B .
- From the graph it is seen that for constant values of I_B ,

I_c is independent on V_{CB} and the curve parallel to the axis V_{CE} .

The output characteristic has 3 regions:-

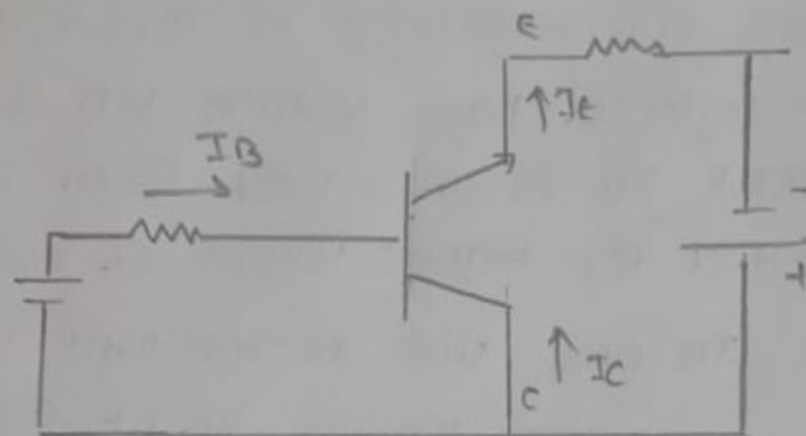
Active region: defined by the biasing arrangement.

Cutoff region: region where the collector current is 0A.

Saturation region: region of the characteristic to the left of $V_{CB}=0$.

q) Common collector (CC) configuration:-

In CC configuration, the collector terminal is common for both input and output circuits. Also known as grounded collector configuration.



$$\text{W.K.T } \Rightarrow I_E = I_B + I_C$$

$$I_C = \alpha I_E + I_{CBO}$$

$$I_E = \alpha I_E + I_{CBO}$$

$$I_E - \alpha I_E = I_B + I_{CBO}$$

$$I_E(1 - \alpha) = I_B + I_{CBO}$$

$$I_E = \frac{1}{1 - \alpha} I_B + \frac{1}{1 - \alpha} I_{CBO}$$

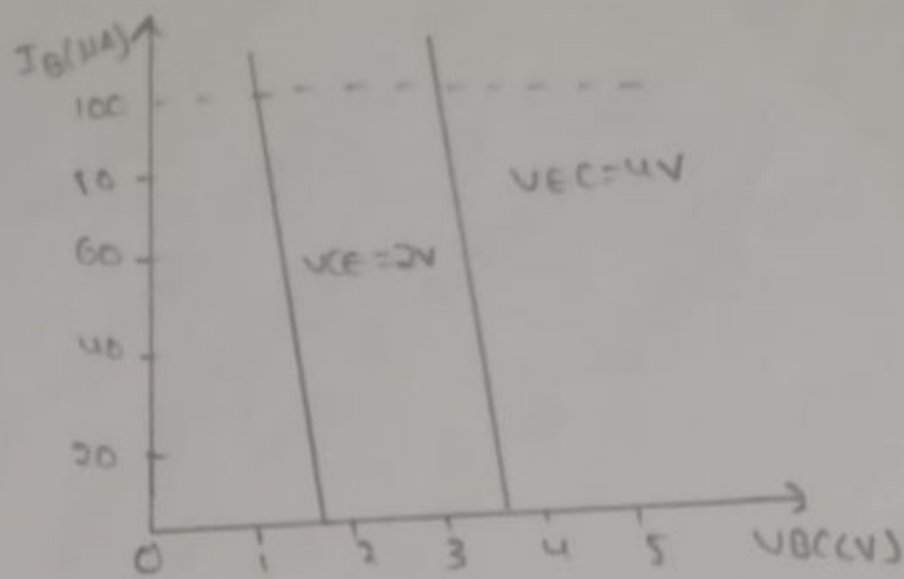
$$I_E = \eta I_B + \eta I_{CBO}$$

$\underbrace{\quad}_{\text{Reverse Saturation}}$

$$\eta = \frac{I_E}{I_B} = \eta = \frac{\Delta I_E}{\Delta I_B}$$

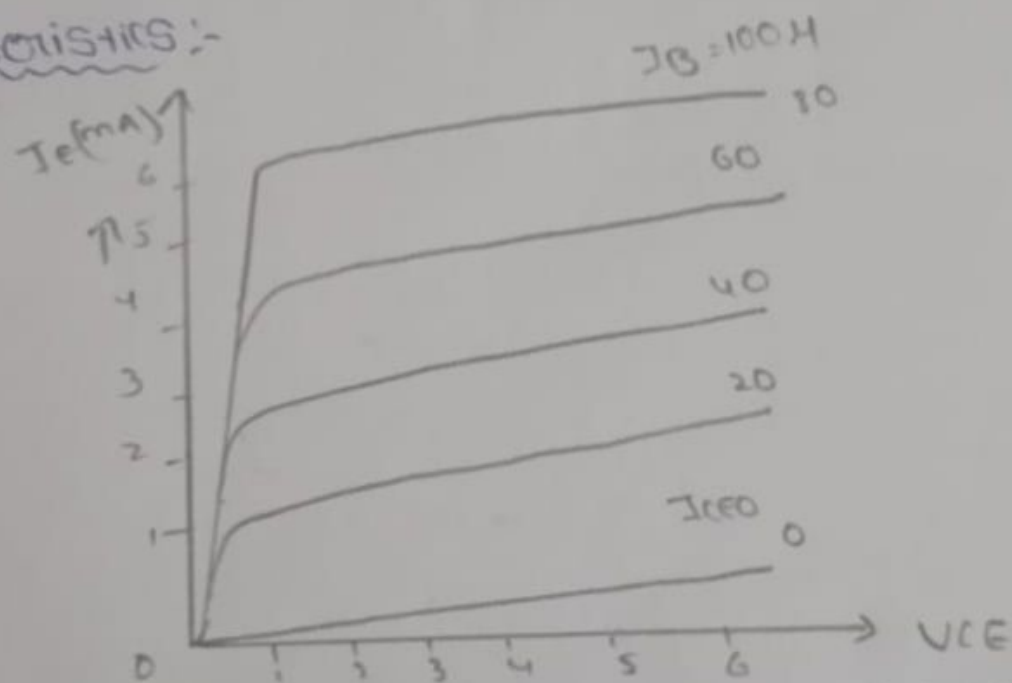
Input characteristics:-

$\Rightarrow$



- It is defined as the characteristic curve drawn b/w input voltage to input current whereas output voltage is constant.
- To determine i/c, the emitter base voltage V_{EB} is kept constant at zero and base current I_B is increased from zero by increasing V_{BE} . This is repeated for higher values of V_{CE} .
- A curve is drawn b/w I_B and V_{EB} at constant V_{CE} .

Output Characteristics:-



- It is defined as the characteristic curve drawn b/w output voltage to output current whereas input current is constant.
- To determine o/c, the base current I_B is kept constant at zero and emitter current I_E is increased from zero by increasing V_{CE} . This is repeated for higher values of I_B .
- From the graph it is seen that for constant value of I_B , I_E is independent of V_{CE} and the curves are parallel to V_{CE} .